Distributed Optimal Secondary Control of Microgrids: A Consensus-Driven Hierarchical Algorithm With Predefined-Time Convergence

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***Abstract*—This brief proposes a hierarchical predefined time optimal secondary control strategy for microgrid systems with external disturbances. Unlike traditional control methods, the key breakthrough of this strategy lies in achieving decoupling between optimization tasks and control tasks: the optimal estimation layer can be separated from other interfering factors and focus on efficient solution of the optimal solution; The control layer independently focuses on dynamic control tasks, effectively avoiding performance limitations caused by task coupling, and providing a new approach for solving optimal control problems in microgrids with unknown nonlinearity and external disturbances. In the optimal estimation layer, a predefined time optimal estimator is design to ensure fast convergence to the optimal solution; At the control layer, a specified performance controller that is not affected by initial values is constructed to ensure that the frequency and voltage of the microgrid converge to the preset accuracy range within a predefined time. Finally, the simulation experiment results demonstrate the effectiveness of the proposed algorithm.**

***Index Terms*—Predefined time, optimal control, microgrid systems, hierarchical control.**

I. Introduction

THE "dual carbon" goal promotes global energy transformation, and the penetration rate of distributed renewable energy is rapidly increasing from the end of the distribution network to the core level. Microgrids have become a research hotspot due to their high reliability, strong flexibility, and good sustainability, which can stabilize energy fluctuations and ensure user demand. In the microgrid control system, secondary control [1], [2] connects primary control with tertiary control [3], which is the core of achieving multi power source coordination and dynamic power balance - primary control stores static errors, while tertiary control is difficult to cope with real-time adjustment and requires refined secondary control support. Therefore, studying the distributed optimal secondary control of microgrids is an important topic.

In the dynamic regulation of microgrids, the fast convergence characteristic is crucial for improving the system's response speed and consensus efficiency, which has become one of the core requirements for ensuring the stable operation of microgrids under high penetration of new energy access. From the perspective of cutting-edge control theory, the key technological paths to achieve this goal focus on advanced nonlinear control methods such as finite time control [4], [5], fixed time control [6], [7], and predefined time control [8], [9]. Ning et al. [10] focused on the application of finite time control in the rapid suppression of frequency and voltage deviations. Another study [11] explored the applicability of fixed time control based on the characteristic of "convergence time independent of initial state" in collaborative regulation of multiple distributed power sources, in order to improve the global consistency of the system. However, predefined time control has become a research hotspot in the field of secondary control of microgrids in recent years due to its ability to accurately define convergence boundaries through parameter presets. Chen et al. [1] proposed a predefined time sliding surface to achieve secondary control of microgrids. However, in current research, there are few studies that integrate optimal control with predefined time control to achieve optimal predefined time quadratic control of microgrids.

The fast convergence characteristic is often accompanied by significant overshoot issues: significant overshoot of key parameters such as frequency and voltage in microgrids may trigger protective device actions, leading to the disconnection of distributed power sources. Therefore, accurately suppressing adverse overshoot and achieving "fast convergence-low overshoot" synergy has become a core challenge that needs to be addressed in the secondary control design of microgrids. The prescribed performance control (PPC) [12], [13], [14], [15] has received widespread attention for its ability to suppress the transient performance of the system. Based on the above discussion, how to realize the optimal predefined-time distributed control for microgrid systems with unknown nonlinearities and external disturbances remains a critical problem that urgently needs to be solved. The research contributions of this brief are summarized as follows: 1) A hierarchical control framework is proposed to decouple the optimal decision-making task from the control task. This framework not only solves the problem of finding the optimal solution for microgrids with disturbances and other factors but

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also achieves the predefined-time control effect. 2) Different from existing finite/fixed optimal results [16], [17], [18], the process of solving the optimal solution in this brief can be completed within a predefined time and does not depend on other parameters of the system. It has stronger flexibility in practical applications and effectively avoids the limitation of application scenarios of traditional methods caused by parameter coupling. 3) In the control layer, compared with existing PPC results [19], [20], the algorithm proposed in this brief not only eliminates the inherent initial condition constraint problem in traditional PPC but also realizes the preset function of convergence time, further improving the adaptability and control performance of the system.

II. Preliminaries and Problem Formulation

A. Systems and Problem Description

According to the research [1], the mathematical model for the frequency and voltage of distributed generator (DG) is as follows:

$$\dot{\omega}_i(t) = u_{\omega_i}(t) + f(\omega_i(t)) + d_{\omega_i}(t), \quad y_{\omega_i}(t) = \omega_i(t) \tag{1}$$

$$\dot{v}_i(t) = u_{v_i}(t) + f(v_i(t)) + d_{v_i}(t), \quad y_{v_i}(t) = v_i(t) \tag{2}$$

where $\omega_i(t)$ and $v_i(t)$ represent the frequency and voltage of the i-th DG respectively. $u_{\omega_i}(t)$ and $u_{v_i}(t)$ denote control inputs. $f(\omega_i(t))$ and $f(v_i(t))$ represents the unknown nonlinear term. $d_{\omega_i}(t)$ and $d_{v_i}(t)$ are bounded disturbance, and $|d_{\omega_i}(t)| \leq \bar{d}$, $|d_{v_i}(t)| < \bar{d}$. $y_{\omega_i}(t)$ and $y_{v_i}(t)$ represent the frequency and voltage output of the i-th DG.

To achieve optimal secondary control of microgrids, this brief divides the control architecture into two layers. Firstly, in the predefined time optimal estimation layer, we design a distributed optimal estimator that only requires local and neighboring information, driving all nodes to converge collaboratively to the global optimal solution within a predefined-time, i.e.

$$\min_{\hat{\omega}_i} \sum_{i=1}^{N} f_i(\hat{\omega}_i) \quad s.t. \quad \hat{\omega}_i = \hat{\omega}_j$$

$$\min_{\hat{v}_i} \sum_{i=1}^{N} f_i(\hat{v}_i) \quad s.t. \quad \hat{v}_i = \hat{v}_j \tag{3}$$

where $\hat{\omega}_i$ and $\hat{v}_i$ are the state of the estimator. Assuming that optimization problems (3) is feasible and represented by $\hat{\omega}^*$ and $\hat{v}^*$ as the optimal solution.

In the second layer, our goal is to design a controller that allows the error e_{ω_i} and e_{v_i} to converge to 0 within a predefined time, i.e. $\lim_{t\to T_s}(\omega_i - \hat{\omega}^*) = 0$ and $\lim_{t\to T_s}(v_i - \hat{v}^*) = 0$.

To achieve the above control objectives, the following assumptions need to be satisfied.

Assumption 1: The communication topology $\mathcal{G}$ is undirected and connected.

Assumption 2: The cost function f_i is twice continuously differentiable and ψ_i - strongly convey with respect to independent variable.

III. Main Results

A. Design of Optimal Predefined Time Estimator

Firstly, the sliding surface of frequency control are given

$$s_{\hat{\omega}_i} = \int_0^t \frac{2c}{p(1-\eta)T_s}\Bigg(\sum_{j=1}^{N} exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) \bullet a_{ij}^{1-p} sig^{1-2p}(\hat{\omega}_i - \hat{\omega}_j)\Bigg)d\tau + \nabla f_i(\hat{\omega}_i) \tag{4}$$

and distributed predefined time optimal estimator

$$\dot{\hat{\omega}}_i = -(H_i(\hat{\omega}_i))^{-1}\Bigg(\frac{1}{2p\eta T_s} exp(\|s_{\hat{\omega}_i}\|^{2p}) sig^{1-2p}(s_{\hat{\omega}_i}) + \frac{2c}{p(1-\eta)T_s}\sum_{j=1}^{N} exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) \bullet a_{ij}^{1-p} sig^{1-p}(\hat{\omega}_i - \hat{\omega}_j)\Bigg) \tag{5}$$

where $H_i(\hat{\omega}_i) = \nabla^2 f_i(\hat{\omega}_i)$, T_s is a convergence constant artificially designed within the physical allowable range. Define $sig(\cdot)^a = |\cdot|^a sgn(\cdot)$, where $sgn(\cdot)$ is the symbolic function.

Theorem 1: Under the conditions of satisfying Assumptions 1 and 2. 1) the sum of local gradients of frequency goes to zero in ηT_s. 2) the proposed optimal signal generator can ensure that the frequency of all DG converge to the optimal solution in a predefined time ηT_s.

Proof: 1) When $t \leq \eta T_s$, taking the derivative of Eq. (4) yields

$$\dot{s}_{\hat{\omega}_i} = -\frac{1}{2p\eta T_s} exp(\|s_{\hat{\omega}_i}\|^{2p}) sig^{1-2p}(s_{\hat{\omega}_i}) \tag{6}$$

Select the following Lyapunov function: $V_\omega = \frac{1}{2}\|s_{\hat{\omega}_i}\|^2$. And, taking the derivative of V_ω and substituting Eq. (6) into it, we can obtain

$$\dot{V}_\omega \leq -\frac{1}{2p\eta T_s} exp(\|s_{\hat{\omega}_i}\|^{2p})\|s_{\hat{\omega}_i}\|^{2-2p} = -\frac{1}{2p\eta T_s} exp(V_\omega^p 2^p) V_\omega^{1-p} \tag{7}$$

Based on the aforementioned results and the predefined time stability theory, we can be concluded that $s_{\hat{\omega}_i}$ can converge to 0 within the predefined time. Next, based on the assumption, we can obtain:

$$\sum_{i=1}^{N}\sum_{j=1}^{N} exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) a_{ij}^{1-p} sig^{1-2p}(\hat{\omega}_i - \hat{\omega}_j) = 0 \tag{8}$$

Further, one has

$$\sum_{i=1}^{N} s_{\hat{\omega}_i} = \sum_{i=1}^{N} \nabla f_i(\hat{\omega}_i) \tag{9}$$

Based on the fact that the sliding mode surface tends to 0 at $t \to \eta T_s$, we can deduce that $\sum_{i=1}^{N} s_{\hat{\omega}_i} = \sum_{i=1}^{N} \nabla f_i(\hat{\omega}_i) = 0$. The first proof has been completed.

2) When $t > \eta T_s$, according to $s_{\hat{\omega}_i} = 0$, one has

$$\dot{\hat{\omega}}_i = -\Bigg(\frac{2c}{p(1-\eta)T_s}\sum_{j=1}^{N} exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p)$$

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$$\bullet\, a_{ij}^{1-p} sig^{1-p}(\hat{\omega}_i - \hat{\omega}_j)\Big)(H_i(\hat{\omega}_i))^{-1} \tag{10}$$

Substituting Eq. (10), it can derived that $d\sum_{i=1}^{N}\nabla f_i(\hat{\omega}_i)/dt = \sum_{i=1}^{N}\nabla^2 f_i(\hat{\omega}_i)\dot{\hat{\omega}}_i = 0$. Because, when $t = \eta T_s$, $\sum_{i=1}^{N}\nabla f_i(\hat{\omega}_i) = 0$, one has $\sum_{i=1}^{N}\nabla f_i(\hat{\omega}_i) = 0, \forall t \geq \eta T_s$. It indicates that the zero-gradient-sum manifold is guaranteed.

Select the following Lyapunov function:

$$V_f = \sum_{i=1}^{N}(f_i(\hat{\omega}_i^*) - f_i(\hat{\omega}_i) - \nabla f_i^T(\hat{\omega}_i)(\hat{\omega}_i^* - \hat{\omega}_i)) \tag{11}$$

According to Assumption 2, it is clear that $V_f \geq \sum_{i=1}^{N}(\psi_2/2)\|\hat{\omega}_i - \hat{\omega}_i^*\|^2$. Let $\tilde{\tilde{\omega}} = [\tilde{\tilde{\omega}}_1^T, \tilde{\tilde{\omega}}_2^T, \ldots, \tilde{\tilde{\omega}}_N^T]$, where $\tilde{\tilde{\omega}}_i = \hat{\omega}_i - \bar{\tilde{\omega}}_i$, with $\bar{\tilde{\omega}}_i = \frac{1}{N}\sum_{k=1}^{N}\hat{\omega}_k$. Then, we has

$$\begin{aligned}
&\sum_{i=1}^{N}(f_i(\bar{\tilde{\omega}}) - f_i(\hat{\omega}_i) - \nabla f_i^T(\hat{\omega}_i)(\bar{\tilde{\omega}} - \hat{\omega}_i)) - V_f \\
&= \sum_{i=1}^{N}(f_i(\bar{\tilde{\omega}}) - f_i(\hat{\omega}^*)) - \left(\sum_{i=1}^{N}\nabla f_i(\hat{\omega}_i)\right)(\bar{\tilde{\omega}} - \hat{\omega}_i) \\
&= \sum_{i=1}^{N}(f_i(\bar{\tilde{\omega}}) - f_i(\hat{\omega}^*)) \geq 0
\end{aligned} \tag{12}$$

Furthermore, the following inequality holds

$$\sum_{i=1}^{N}(f_i(\bar{\tilde{\omega}}) - f_i(\hat{\omega}_i) - \nabla f_i^T(\hat{\omega}_i)(\bar{\tilde{\omega}} - \hat{\omega}_i)) \leq \frac{\bar{\psi}}{2}\|\tilde{\tilde{\omega}}_i\|^2 \tag{13}$$

So, one has

$$V_f \leq \frac{\bar{\psi}}{2}\|\tilde{\tilde{\omega}}_i\|^2 \tag{14}$$

Differentiate Eq. (11) and substitute Eq. (13) into $\dot{V}_f$ to obtain

$$\begin{aligned}
\dot{V}_f = &\sum_{i=1}^{N}(\hat{\omega}_i - \hat{\omega}_i^*)^T\nabla^2 f_i(\hat{\omega}_i)\dot{\hat{\omega}}_i \\
&- \frac{2c}{p(1-\eta)T_s}\sum_{i=1}^{N}\sum_{j=1}^{N} exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) \\
&\bullet\, a_{ij}^{1-p}(\hat{\omega}_i - \hat{\omega}_i^*)^T sig^{1-2p}(\hat{\omega}_i - \hat{\omega}_j)
\end{aligned} \tag{15}$$

And because

$$\sum_{i=1}^{N}\sum_{j=1}^{N} exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p)a_{ij}^{1-p}(\hat{\omega}_i^*)^T sig^{1-2p}(\hat{\omega}_i - \hat{\omega}_j) = 0$$

According to the above result, $\dot{V}_f$ can be rewritten as

$$\begin{aligned}
\dot{V}_f &= -\frac{2c}{p(1-\eta)T_s}\sum_{i=1}^{N}\sum_{j=1}^{N}(exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) \\
&\bullet\, a_{ij}^{1-p}\|\hat{\omega}_i - \hat{\omega}_j\|_{2-2p}^{2-2p}) \\
&\leq -\frac{2c}{p(1-\eta)T_s}\sum_{i=1}^{N}\sum_{j=1}^{N}(exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) \\
&\bullet\, (a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^{1-p})
\end{aligned} \tag{16}$$

By using Jensen's inequality and $f(m) = exp(m^{2p})m^{2(1-p)}$, we have

$$\begin{aligned}
&\frac{1}{N(N-1)}\sum_{i=1}^{N}\sum_{j=1}^{N}(exp((a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^p) \\
&(a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2)^{1-p}) \\
&\geq exp\left(\left(\frac{\sum_{i=1}^{N}\sum_{j=1}^{N}a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2}{N(N-1)}\right)^p\right) \\
&exp\left(\frac{\sum_{i=1}^{N}\sum_{j=1}^{N}a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2}{N(N-1)}\right)^{1-p}
\end{aligned} \tag{17}$$

And since the graph is undirected and connected, $1^T\tilde{\tilde{\omega}} = 0$, $\lambda_2(L) = \lambda_2(L \otimes I_n)$, one has

$$\sum_{i=1}^{N}\sum_{j=1}^{N}a_{ij}\|\hat{\omega}_i - \hat{\omega}_j\|^2 = 2\tilde{\tilde{\omega}}^T(L \otimes I_n)\tilde{\tilde{\omega}} \geq 2\lambda_2(L)\|\tilde{\tilde{\omega}}\|^2 \tag{18}$$

Finally, $\dot{V}_f$ can be rewritten as

$$\begin{aligned}
\dot{V}_f \leq &-\frac{cN(N-1)}{p(1-\eta)T_s}\left(exp\left(\left(\frac{2\lambda_2(L)\|\tilde{\tilde{\omega}}\|^2}{N(n-1)}\right)^2\right)\right. \\
&\left.\bullet\left(\frac{2\lambda_2(L)\|\tilde{\tilde{\omega}}\|}{N(N-1)}\right)^{1-p}\right)
\end{aligned} \tag{19}$$

From Eq. (14) and Eq. (19), one has

$$\begin{aligned}
\dot{V}_f &\leq -\beta_1\left(\frac{4\lambda_2(L)}{N(n-1)\bar{\psi}}\right)^{1-p}exp\left(\left(\frac{4\lambda_2(L)v_f}{N(n-1)\bar{\psi}}\right)^p\right)V_f^{1-p} \\
&\leq -\beta_2\left(\frac{4\lambda_2(L)}{N(n-1)\bar{\psi}}\right)^{1-p}exp\left(\left(\frac{4\lambda_2(L)v_f}{N(n-1)\bar{\psi}}\right)^p\right)V_f^{1-p}
\end{aligned} \tag{20}$$

where $\beta_1 = \frac{cN(N-1)}{p(1-\eta)T_s}, \beta_2 = \frac{1}{p(1-\eta)T_s}$. At this point, with the help of optimal conditions and predefined time theory [21], we can obtain $\lim_{t\to T_s}\hat{\omega}_i(t) \to \hat{\omega}_i^*$. □

Similar to the estimator results for frequency secondary control, those of the estimator for voltage secondary control will be provided directly.

Sliding surface of voltage secondary control:

$$\begin{aligned}
s_{\hat{v}_i} = \int_0^t \frac{2c}{p(1-\eta)T_s}\Bigg(&\sum_{j=1}^{N}exp((a_{ij}\|\hat{v}_i - \hat{v}_j\|^2)^p) \\
&\bullet\, a_{ij}^{1-p}sig^{1-2p}(\hat{v}_i - \hat{v}_j)\Bigg) + \nabla f_i(\hat{v}_i)
\end{aligned} \tag{21}$$

and optimal estimator is designed to

$$\begin{aligned}
\dot{\hat{v}}_i = -(H_i(\hat{v}_i))^{-1}\Bigg(&\frac{1}{2p\eta T_s}exp(\|s_{\hat{v}_i}\|^{2p})sig^{1-2p}(s_{\hat{v}_i}) \\
&+ \frac{2c}{p(1-\eta)T_s}\sum_{j=1}^{N}exp((a_{ij}\|\hat{v}_i - \hat{v}_j\|^2)^p) \\
&\bullet\, a_{ij}^{1-p}sig^{1-p}(\hat{v}_i - \hat{v}_j)\Bigg)
\end{aligned} \tag{22}$$

Based on the design of the optimal trajectory mentioned above, the secondary control results of voltage are summarized as *Theorem 2*.

Theorem 2: Under the conditions of satisfying Assumptions 1 and 2. 1) the sum of local gradients of voltage goes to zero in ηT_s. 2) the proposed optimal signal generator can ensure that the voltage of all DG converge to the optimal solution in a predefined time ηT_s.

Due to space limitations, the stability proofs of frequency control estimator and voltage control estimator are similar.

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Therefore, we will only provide stability analysis of the frequency control estimator. The stability analysis of voltage estimator can be obtained in a similar way.

B. Design of Control

In this section, to achieve the goal of predefined time control, we first present the following nonlinear transformation function:

$$\vartheta(t) = \begin{cases} \dfrac{t^2}{(T_o - t)^2 + \mu t^2}, & 0 \le t < T_o \\ \dfrac{1}{\mu}, & t \ge T_o \end{cases} \tag{23}$$

where T_o is an adjustable convergence time parameter within the allowed range of physics. μ is an adjustable convergence accuracy parameter.

It should be pointed out that the above nonlinear transformation function is not fixed and invariant; As long as the following three properties are met. 1) $\vartheta(0) = 0$; 2) It strictly monotonically increases within the interval $[0, T_o]$ and maintains its maximum value unchanged after reaching time T_o. 3) It is continuously differentiable and its derivative $\dot{\vartheta}(t)$ is bounded, i.e. $|\dot{\vartheta}(t)| \le \bar{\vartheta}$ with $\bar{\vartheta}$ being a positive constant.

Based on the nonlinear transformation function mentioned above, construct the following error variables:

$$e_{\omega_i} = \omega_i - \hat{\omega}_i, \quad z_{\omega_i} = \vartheta e_{\omega_i} \tag{24}$$

Then based on Eq. (1), we can obtain

$$\dot{e}_{\omega_i} = u_{\omega_i}(t) + f(\omega_i(t)) + d_{\omega_i}(t) - \dot{\hat{\omega}}_i \tag{25}$$

And because $z_{\omega_i} = \vartheta e_{\omega_i}$, one has

$$\dot{z}_{\omega_i} = \dot{\vartheta} e_{\omega_i} + \vartheta(u_{\omega_i}(t) + f(\omega_i(t)) + d_{\omega_i}(t) - \dot{\hat{\omega}}_i) \tag{26}$$

Because the disturbance $d_{\omega_i}(t)$ in the system is unknown, we need the following disturbance observer to estimate it. Let

$$\begin{aligned} \hat{d}_{\omega_i}(t) &= p_i(t) - \iota\omega_i(t) \\ \dot{p}_i(t) &= \iota(p_i(t) - \iota\omega_i(t)) + \iota(f(\omega_i(t)) + u_{\omega_i})) \end{aligned} \tag{27}$$

where $\hat{d}_{\omega_i}(t)$ is estimation of disturbance; $p_i(t)$ is the state of the disturbance observer; $\tilde{d}_{\omega_i} = d_{\omega_i} - \hat{d}_{\omega_i}$ is bounded error, and defined by $\bar{\tilde{d}}_{\omega_i} = \sup_{t>0} |\tilde{d}_{\omega_i}|$.

According to the disturbance observer, we can obtain:

$$\begin{aligned} \dot{\hat{d}}_{\omega_i}(t) &= \dot{p}_i(t) - \iota\omega_i(t) \\ &= \iota(p_i(t) - \iota\omega_i(t)) + \iota(u_{\omega_i} + f(\omega_i)) \\ &\quad - \iota(u_{\omega_i} + d_{\omega_i}(t) + f(\omega_i)) = -\iota\tilde{d}_{\omega_i}(t) \end{aligned} \tag{28}$$

So, from Eq. (28), one has

$$\dot{\tilde{d}}_{\omega_i}(t) = \dot{d}_{\omega_i}(t) - \dot{\hat{d}}_{\omega_i}(t) = \dot{d}_{\omega_i}(t) + \iota\tilde{d}_{\omega_i}(t) \tag{29}$$

Finally, based on the tracking error and disturbance observer, the following controller is designed

$$u_{\omega_i} = -k_\omega e_{\omega_i} - W_\omega^{*T}\varphi(\omega_i) - \hat{d}_{\omega_i} + \dot{\hat{\omega}}_i - \varrho_\omega(\cdot) \tag{30}$$

where W_ω^* and $\varphi(\omega_i)$ represent the ideal weights and radial basis functions of the RBF neural network, respectively. i.e. $f(\omega_i) = W_\omega^{*T}\varphi(\omega_i) + \sigma(\omega_i)$, where $\sigma(\omega_i)$ is an approximation error and satisfies $|\sigma(\omega_i)| \le \bar{\sigma}$. $\varrho_\omega(\cdot) = \frac{\alpha_1 e_{\omega_i}^3}{4(b^2 - z_{\omega_i}^2)} + \frac{z_{\omega_i}\vartheta}{2(b^2 - z_{\omega_i}^2)}$.

Theorem 3: Under the designed controller Eq.(30), the tracking error can converge to the given tracking accuracy within a predefined time.

Proof: Choose Lyapunov function $V_1 = \frac{1}{2}\ln\frac{b^2}{b^2 - z_{\omega_i}^2}$ and taking its derivative, one has

$$\dot{V}_1 \le \frac{z_{\omega_i}\dot{\vartheta} e_{\omega_i}}{b^2 - z_{\omega_i}^2} + \frac{z_{\omega_i}\vartheta}{b^2 - z_{\omega_i}^2}(u_{\omega_i} + f(\omega_i) + d_{\omega_i} - \hat{\omega}_i) \tag{31}$$

According to Yang's inequality, one has

$$\begin{aligned} \frac{z_{\omega_i}\dot{\vartheta} e_{\omega_i}}{b^2 - z_{\omega_i}^2} &\le \frac{\alpha_1 z_{\omega_i}^2 e_{\omega_i}^2}{4(b^2 - z_{\omega_i}^2)} + \frac{\bar{\vartheta}}{\alpha_1} \\ \frac{z_{\omega_i}\vartheta}{b^2 - z_{\omega_i}^2}(\sigma + \tilde{d}_{\omega_i}) &\le \frac{2\alpha_1 z_{\omega_i}^2 \vartheta^2}{b^2 - z_{\omega_i}^2} + \frac{\bar{\sigma} + \bar{\tilde{d}}_{\omega_i}}{\alpha_1} \end{aligned} \tag{32}$$

Substituting the controller (30) and Eq. (32) into Eq. (31), $\dot{V}_1$ can be rewritten as

$$\dot{V}_1 \le \frac{k_\omega e_{\omega_i}^2}{b^2 - z_{\omega_i}^2} + \Lambda_\omega \le -k_\omega V_1 + \Lambda_\omega \tag{33}$$

where $\Lambda_\omega = \frac{\bar{\vartheta}}{\alpha_1} + \frac{\bar{\sigma} + \bar{\tilde{d}}_{\omega_i}}{\alpha_1}$. Furthermore, based on the stability analysis above, it can be concluded that $|z_{\omega_i}| < b, \forall t > 0$, and according to the nonlinear transformation function, it can be obtained

$$|e_\omega| < \vartheta^{-1} b = \begin{cases} \left(\frac{T_o - t}{t}\right)^2 + \mu b, & 0 < t < T_o \\ \mu b, & t \ge T_o \end{cases} \tag{34}$$

As shown in Eq. (34), the tracking error decays at a rate not lower than rate $((T_o - T)/t)^2$ a within the interval $(0, T_o)$ and eventually converges to within the preset accuracy $(-\mu b, \mu b)$. At this point, the stability proof has been completed. □

Next, we will briefly present the controller structure for voltage secondary control. Similar to (24), provide similar errors

$$e_{v_i} = v_i - \hat{v}_i, \quad z_{v_i} = \vartheta e_{v_i} \tag{35}$$

Then, choose the Lyapunov function $V_2 = \frac{1}{2}\ln\frac{b^2}{b^2 - z_{v_i}^2}$, and the controller is designed as

$$u_{v_i} = -k_v e_{v_i} - W_v^{*T}\varphi(v_i) - \hat{d}_{v_i} + \hat{v}_i - \varrho_v(\cdot) \tag{36}$$

The design of relevant parameters has been provided in the frequency secondary control section, and will not be repeated here; Therefore, the conclusion of voltage secondary control can be summarized as *Theorem 4*.

Theorem 4: Under the designed controller Eq. (24), the tracking error of voltage can converge to the given tracking accuracy within a predefined time.

The stability analysis of *Theorem 4* is similar to that of frequency control, so it will not be explained in detail.

IV. Simulation

To verify the effectiveness of the proposed hierarchical control framework, numerical simulations were conducted in this section. The initial values of the frequency and voltage of the estimation layer are selected as $\hat{\omega}_i = [49.5, 50.5, 50.9]^T$, $\hat{v}_i = [385, 375, 383]^T$. The initial values of frequency and voltage for the control layer are selected as $\omega_i = [48, 51, 53]^T, v_i = [360, 370, 390]^T$. The cost function is

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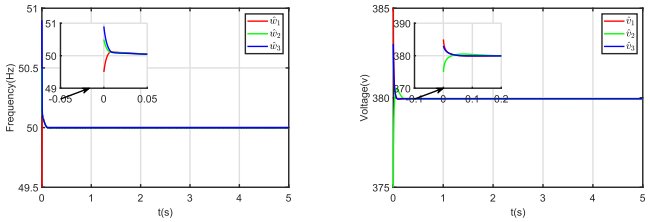


Fig. 1. The trajectory curve of the estimate layer.

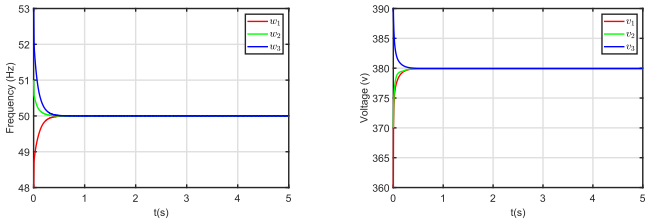


Fig. 2. The trajectory curve of the frequency and voltage.

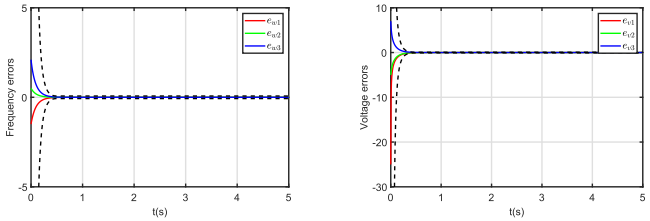


Fig. 3. The trajectory curve of e_{wi} and e_{vi}.

selected as $f_i(\hat{\omega}_i) = (\hat{\omega}_i - 50)^2; f_i(\hat{v}_i) = (\hat{v}_i - 380)^2$. The parameter has been selected as $p = 0.3, \eta = 0.4, \mu = 0.08, T_s = 1s, T_p = 0.5$. The disturbance is selected as $d_{\omega_i}(t) = d_{v_i}(t) = 2\sin(t)$. The nonlinear function is selected as $f_i(\omega_i) = 0.5\sin(\omega_i(t)); f_i(v_i) = 0.5\cos(v_i(t))$.

Fig. 1 shows that the estimator can accurately converge to the optimal solution within the predefined time; Fig. 2 shows that the control layer can effectively control frequency and voltage within a predefined time; Fig. 3 clearly demonstrates that the controller can converge the frequency and voltage to the preset accuracy without violating performance constraints.

V. Conclusion

This brief proposes a hierarchical predefined time optimal secondary control strategy for microgrid systems with external disturbances and unknown nonlinear functions. The optimal estimation layer can be detached from external disturbance factors and focus on efficient solution of the optimal solution. The design of the predefined time optimal estimator ensures fast convergence of the optimal solution; The control layer independently focuses on dynamic control tasks, and the constructed performance controller that is not affected by initial values can accurately ensure that the frequency and voltage of the microgrid converge to the preset accuracy range within a predefined time, providing a new technical path for optimal control problems of microgrids with unknown nonlinearity and external disturbances. In the future, we will combine reinforcement learning control strategies to further optimize the controller.

References

[1] X. Chen, S. Hu, X. Xie, and J. Qiu, "Consensus-based distributed secondary control of microgrids: A pre-assigned time sliding mode approach," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 12, pp. 2525–2527, Dec. 2024.

[2] H. Yang, C. Deng, X. Xie, and L. Ding, "Distributed resilient secondary control for AC microgrid under FDI attacks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 7, pp. 2570–2574, Jul. 2023.

[3] J. M. Guerrero, J. C. Vasquez, J. Matas, L. G. de Vicuna, and M. Castilla, "Hierarchical control of droop-controlled AC and DC microgrids—A general approach toward standardization," *IEEE Trans. Ind. Electron.*, vol. 58, no. 1, pp. 158–172, Jan. 2011.

[4] C. Fu, C. Zhang, G. Zhang, and Q. Su, "Finite-time adaptive fuzzy control for three-phase PWM rectifiers with improved output performance," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 8, pp. 3044–3048, Aug. 2023.

[5] H. Wang et al., "Current-constrained finite-time control scheme for speed regulation of PMSM systems with unmatched disturbances," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 16877–16887, 2025.

[6] T. Wei and X. Li, "Fixed-time and predefined-time stability of impulsive systems," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 4, pp. 1086–1089, Apr. 2023.

[7] Z. Zuo, Q.-L. Han, B. Ning, X. Ge, and X.-M. Zhang, "An overview of recent advances in fixed-time cooperative control of multiagent systems," *IEEE Trans. Ind. Informat.*, vol. 14, no. 6, pp. 2322–2334, Jun. 2018.

[8] R. Zhang, G. Guo, and Z.-D. Zhou, "Balance of communication and convergence: Predefined-time distributed optimization based on zero-gradient-sum," *IEEE Trans. Cybern.*, vol. 55, no. 2, pp. 661–671, Feb. 2025.

[9] C. Jia, X. Liu, X. Li, and J. Xu, "A new adaptive predefined time sliding mode control for nonlinear systems," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 4, pp. 2094–2098, Apr. 2024.

[10] B. Ning, Q.-L. Han, and L. Ding, "Distributed finite-time secondary frequency and voltage control for islanded microgrids with communication delays and switching topologies," *IEEE Trans. Cybern.*, vol. 51, no. 8, pp. 3988–3999, Aug. 2021.

[11] Y. Zeng et al., "Fixed-time secondary controller for rapid state-of-charge balancing and flexible bus voltage regulation in DC microgrids," *IEEE Trans. Power Syst.*, vol. 39, no. 3, pp. 5393–5407, May 2024.

[12] P. Ning, D. Mu, and C. Hua, "Global prescribed-time tracking control mobile robot systems with performance constrains," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 1, pp. 181–185, Jan. 2024.

[13] N. Wang and Y. Wang, "Fuzzy adaptive quantized tracking control of switched high-order nonlinear systems: A new fixed-time prescribed performance method," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 7, pp. 3279–3283, Jul. 2022.

[14] W. Lai, Y. Li, and Z. Lin, "Prescribed performance control for a nonlinear system with actuator saturation via anti-windup design," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 11, pp. 7281–7294, Nov. 2025, doi: 10.1109/TCSI.2025.3566467.

[15] Y. Zhang, Z. Liu, F. Wang, and Z. Chen, "Distributed power allocation scheme with prescribed performance and intermittent dynamics for BESSs with discharge rate constraints in microgrids," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 55, no. 3, pp. 1948–1959, Mar. 2025.

[16] H. Liu, W. X. Zheng, and W. Yu, "Continuous-time algorithm based on finite-time consensus for distributed constrained convex optimization," *IEEE Trans. Autom. Control*, vol. 67, no. 5, pp. 2552–2559, May 2022.

[17] S. Sun, Y. Zhang, P. Lin, W. Ren, and J. A. Farrell, "Distributed time-varying optimization with state-dependent gains: Algorithms and experiments," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 1, pp. 416–425, Jan. 2022.

[18] G. Guo, Z.-D. Zhou, and R. Zhang, "Distributed fixed-time optimization with time-varying cost: Zero-gradient-sum scheme," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 6, pp. 3086–3090, Jun. 2024.

[19] Y. Hou and B. Cheng, "Event-based H_∞ consensus of double-integrator multi-agent systems: A prescribed performance control approach," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 4, pp. 2129–2133, Apr. 2024.

[20] C. Ding, S. Ding, and K. Mei, "Adaptive prescribed-time SOSM controller design for nonlinear systems with prescribed performance," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 3, pp. 1311–1315, Mar. 2024.

[21] J. D. Sánchez-Torres, M. Defoort, and A. J. Muñoz-Vázquez, "Predefined-time stabilisation of a class of nonholonomic systems," *Int. J. Control*, vol. 93, no. 12, pp. 2941–2948, Dec. 2020.

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